22. Einstein Ring

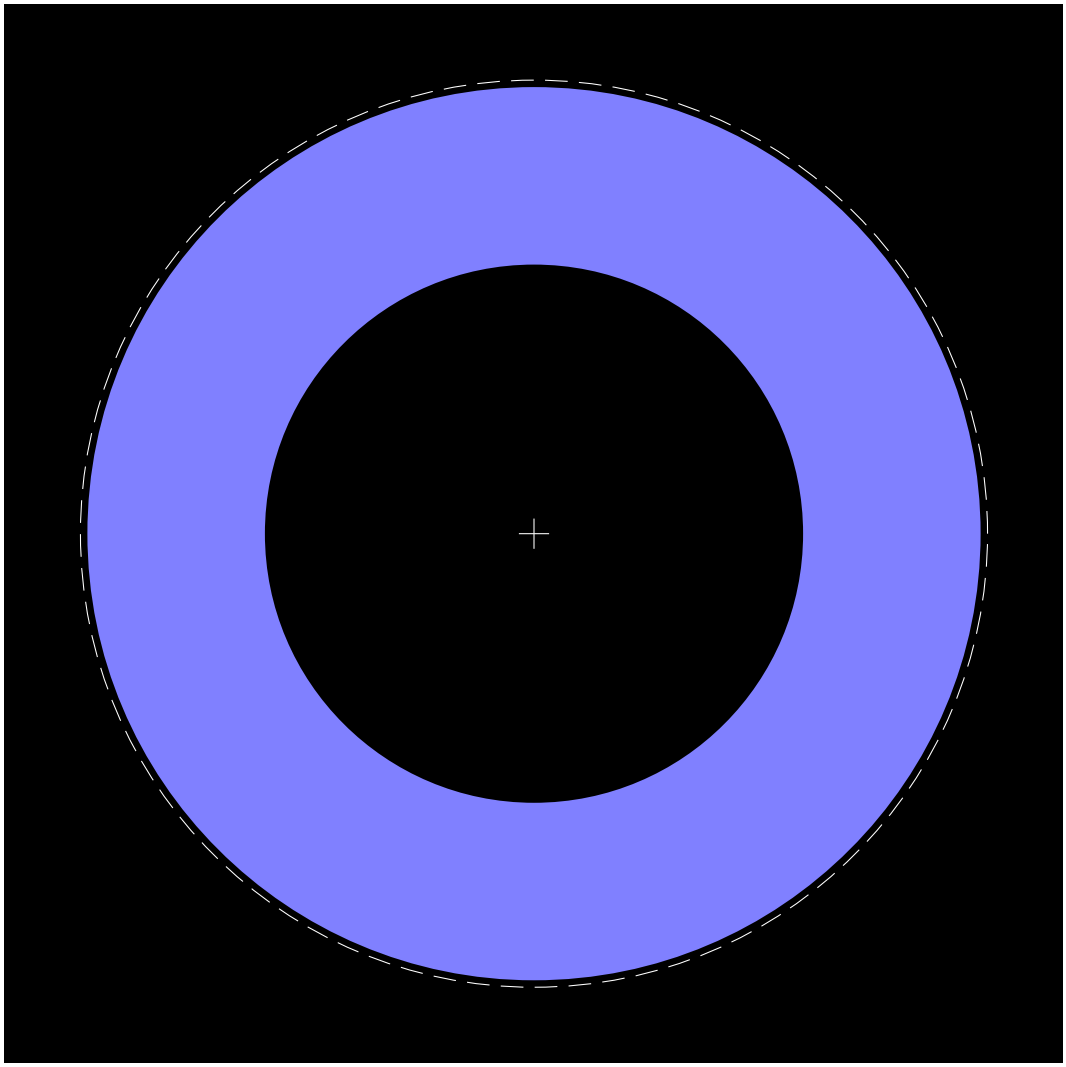
A transparent unit sphere has a small hemispherical indent in its surface, located at $\hat{z}$. UV light shines on the indent from the $\hat{z}$ direction. The sphere is doped with fluorescent dye that has a chance to absorb UV light and reemit it as blue light in a random direction. The concentration of dye is small, so a UV photon typically travels a very long distance before being absorbed. Assume that light is either totally reflected or transmitted at interfaces.

A camera far away from the sphere in the $\hat{z}$ direction takes the following image; the center and boundary of the sphere are marked. If the sphere's indices of refraction for blue and UV light are n_b and n_u respectively:

(a) Find n_b. [6.0 pts]

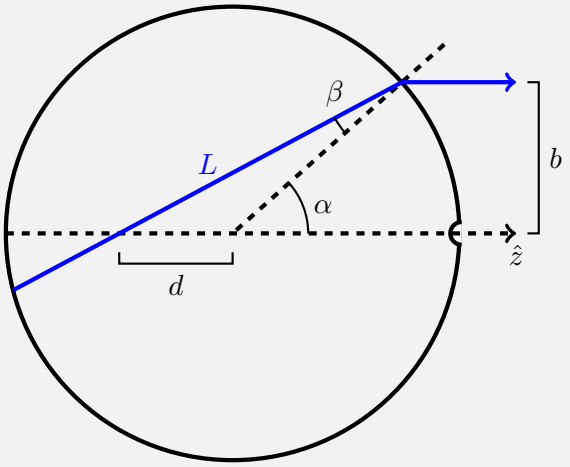
(b) Find n_u. [5.0 pts]

(c) For which values (n_b, n_u) does a similar ring appear? [4.0 pts]

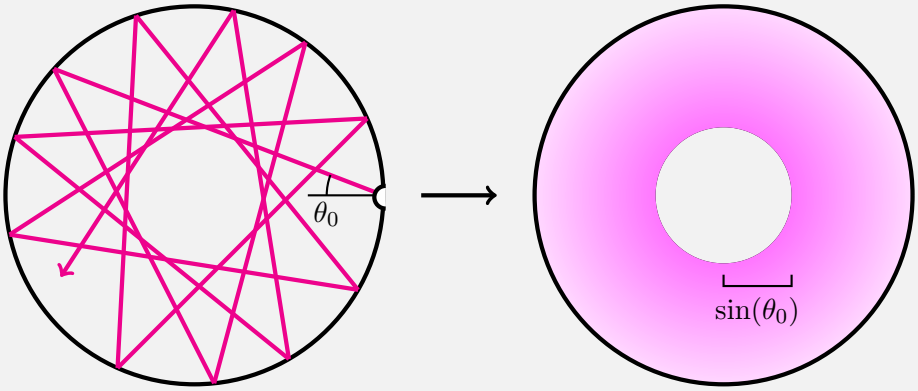


Solution 22: (a-b) Note that all blue photons in the image exit the sphere directly after being emitted, as reflected blue photons are trapped and we are ignoring partial reflections. Thus, to find the intensity $I(b)$ at an image point with impact parameter $b < 1$, we integrate over the possible origins for a trajectory leading to that point, which lie along the refracted segment L. Since the emitted photon's direction is uniform, we do not need to know the absorbed photon's direction, so it suffices to know the distribution of absorption locations $P(\vec{r})$.

$$I(b) \propto \int_L P(\vec{r})\, d\ell$$



To find $P(\vec{r})$, note that UV photons which leave the sphere are unlikely to be absorbed, so it suffices to consider UV photons trapped inside the sphere by reflection, which must originate at the hemisphere. Consider the trajectory of a single UV photon originating at azimuth ϕ and polar angle θ_0, which is trapped if $\sin(\theta_0) > 1/n_u$. Generically, θ_0 is irrational, so the final polar angle θ is approximately uniformly distributed and we can take $\vec{r}$ to be sampled from the plane of the trajectory with some probability distribution $P(r, \theta \mid \theta_0, \phi) = f(r)$. Note that $f(r)$ is nonzero for $r \in (\sin(\theta_0), 1)$.



Then, given a UV photon emitted at azimuth ϕ, we see that $\vec{r}$ has distribution $P(r, \theta \mid \phi) = \int_0^{\arcsin(1/n_u)} P(r, \theta \mid \theta_0, \phi) P(\theta_0 \mid \phi)\, d\theta_0 = g(r)$, which is nonzero for $r \in (1/n_u, 1)$. Finally, the unconditional distribution is $P(r, \theta, \phi) = P(r, \theta \mid \phi) P(\phi) = \frac{1}{2\pi} g(r)$, giving $P(\vec{r}) = \frac{1}{r^2 \sin(\theta)} P(r, \theta, \phi) = \frac{g(r)}{2\pi r^2 \sin(\theta)}$. We see that $P(\vec{r})$ diverges if $\theta = 0, \pi$ and $r \in (1/n_u, 1)$; physically, this is because the trajectories of all UV photons intersect these segments, with the divergence being cut off by the finite size of the source.

Thus, $I(b)$ is large if L intersects the locus of divergence, i.e. $d \in (1/n_u, 1)$. Note that:

$$\begin{aligned} d &= \frac{\sin(\alpha)}{\tan(\alpha-\beta)} - \cos(\alpha) \\ &= \frac{\sin(\arcsin(b))}{\tan(\arcsin(b) - \arcsin(b/n_b))} - \cos(\arcsin(b)) \\ &= b \Bigg/ \left(\frac{\frac{b}{\sqrt{1-b^2}} - \frac{b/n_b}{\sqrt{1-(b/n_b)^2}}}{1 + \frac{b}{\sqrt{1-b^2}} \frac{b/n_b}{\sqrt{1-(b/n_b)^2}}} \right) - \sqrt{1-b^2} \\ &= \frac{\sqrt{(1-b^2)(n_b^2-b^2)} + b^2}{\sqrt{n_b^2-b^2} - \sqrt{1-b^2}} - \sqrt{1-b^2} \\ &= \frac{1}{\sqrt{n_b^2-b^2} - \sqrt{1-b^2}} \end{aligned}$$

This is a decreasing function of b, explaining the appearance of the ring. In particular, the inner radius $b_1 = 0.593$ of the ring corresponds to $d = 1$ and the outer radius $b_2 = 0.984$ corresponds to $d = 1/n_u$, giving:

$$n_b = \sqrt{2\left(1 + \sqrt{1-b_1^2}\right)} = \boxed{1.900}, \quad n_u = \sqrt{n_b^2 - b_2^2} - \sqrt{1-b_2^2} = \boxed{1.45}$$

(c) Note that $d(0) = \dfrac{1}{n_b - 1}, d(1) = \dfrac{1}{\sqrt{n_b^2 - 1}}$. For a ring (and not a disk) to appear, we need $d(0) > 1, d(1) < 1$, so $\boxed{n_b \in (\sqrt{2}, 2)}$. This is not the only constraint: we must also ensure that some UV light remains trapped in the sphere. For this, consider a UV ray with initial impact parameter $0 < b' < 1$ on the hemisphere; we calculate $\theta_0 = \arcsin(b) - \arcsin(n_u/b')$. We see that $\dfrac{d\theta_0}{db'} = \dfrac{1}{\sqrt{1-b'^2}} - \dfrac{1}{\sqrt{n_u^2 - b'^2}} > 0$, so the maximum value of θ_0 is $\theta_m = \pi/2 - \arcsin(1/n_u)$. We need $\theta_m > \arcsin(1/n_u)$, implying $\boxed{n_u > \sqrt{2}}$.

Remark: One might expect that $I(b) = \displaystyle\int_L S(b, \vec{r}) P(\vec{r})\, d\ell$ where $S(b, \vec{r})$ is a geometric factor accounting for effects such as magnification. In fact, due to a principle called the *conservation of etendue*, $S(b, \vec{r})$ is constant. This knowledge is not necessary to solve the problem.

Remark: In order for the ring to be the dominant effect observed, we should check that the $\hat{z}$ axis is the *only* place where $P(\vec{r})$ diverges; in particular, $P(r, \theta \mid \phi)$ must be bounded. To see this, note that $P(r, \theta \mid \phi) = \displaystyle\int_{1/n_u}^{\arcsin(\theta_m)} P(r, \theta \mid \theta_0, \phi) P(\sin(\theta_0) \mid \phi)\, d\sin(\theta_0)$. One can easily calculate $P(r, \theta \mid \theta_0, \phi) = \dfrac{1}{\cos(\theta_0)} \dfrac{r}{\sqrt{r^2 - \sin^2(\theta_0)}}$ for $r \in (\sin(\theta_0), 1)$. Furthermore, $P(\sin(\theta_0) \mid \phi) = P(b \mid \phi) / \dfrac{d\sin(\theta_0)}{db} = \dfrac{b}{2} \dfrac{1}{\cos(\theta_0) \frac{d\theta_0}{db}}$. Because $\cos(\theta_0) > \cos(\theta_m)$ and $\dfrac{d\theta_0}{db} > 1 - 1/n_u$ are bounded away from 0, we can write $P(r, \theta \mid \phi) < C \displaystyle\int_{1/n_u}^{\arcsin(\theta_m)} \frac{r}{\sqrt{r^2 - \sin^2(\theta_0)}}\, d\sin(\theta_0)$, which is clearly bounded.